Neuromodulated Spike-Timing-Dependent Plasticity

Model Specification, Reward Regimes, and Numerical Analysis

1 Microscopic Foundation: Neural Dynamics

We consider a minimal circuit: one presynaptic neuron connected to one postsynaptic neuron through a unidirectional synapse of weight w.

1.1 Membrane Potential (LIF)

The postsynaptic neuron is a leaky integrate-and-fire (LIF) unit (Dayan & Abbott, 2001; Gerstner et al., 2014). Its membrane potential $V(t)$ obeys

$$\tau_m \frac{dV}{dt} = -(V - E_L) + R_m I_{\text{syn}(t)} \tag{1}$$

where $\tau_m = R_m C_m$ is the membrane time constant, E_L the resting potential, and I_{syn} the synaptic current. A spike fires at time t^k_{post} when $V(t) = \theta$ with $dV/dt > 0$; the membrane resets to V_{reset} and integration is suspended for a refractory period τ_{ref}.

1.2 Spike Trains

The spike trains of both neurons are sums of Dirac deltas:

$$\rho_{\text{pre}(t)} = \sum_k \delta\big(t - t^k_{\text{pre}}\big), \quad \rho_{\text{post}(t)} = \sum_k \delta\big(t - t^k_{\text{post}}\big). \tag{2}$$

Presynaptic times $\{t^k_{\text{pre}}\}$ are given (regular at rate r_{pre} Hz); postsynaptic times are determined by Equation 1.

1.3 Synaptic Interaction

The synaptic current is the presynaptic spike train filtered through an exponential kernel and scaled by w:

$$I_{\text{syn}(t)} = w \int_{-\infty}^{t} \alpha(t-s)\rho_{\text{pre}(s)} ds, \quad \alpha(t) = \tau_s^{-1} e^{-t/\tau_s} \Theta(t). \tag{3}$$

In ODE form: $dI_s/dt = -I_s/\tau_s$, with $I_s \leftarrow I_s + 1$ at each presynaptic spike.

2 Three-Factor Plasticity Model

The plasticity rule belongs to the class of *three-factor learning rules* (Frémaux & Gerstner, 2016; Izhikevich, 2007). Standard STDP depends on pre–post spike correlations; three-factor rules gate this local signal with a global neuromodulatory factor.

2.1 STDP Traces

A presynaptic trace x_{pre} and postsynaptic trace y_{post} capture recent spike history (Bi & Poo, 1998):

$$\tau_+ \frac{dx_{\text{pre}}}{dt} = -x_{\text{pre}} + \rho_{\text{pre}(t)}, \quad \tau_- \frac{dy_{\text{post}}}{dt} = -y_{\text{post}} + \rho_{\text{post}(t)}. \tag{4}$$

Experimental measurements place $\tau_+ \approx \tau_- \approx 20$ ms.

2.2 Eligibility Trace

Coincident pre- and post-spikes create a temporary memory, the *eligibility trace* $E(t)$ (Gerstner et al., 2018; Izhikevich, 2007), bridging the gap between millisecond spike timing and delayed reward:

$$\tau_e \frac{dE}{dt} = -E + \underbrace{\eta_+ (w_{\max} - w) x_{\text{pre}} \rho_{\text{post}}}_{\text{LTP}} - \underbrace{\eta_- w y_{\text{post}} \rho_{\text{pre}}}_{\text{LTD}} \tag{5}$$

The soft-bound factors $(w_{\max} - w)$ and w implement *multiplicative STDP* (Song et al., 2000), preventing unbounded weight growth.

2.3 Firing-Rate Estimation

Smooth rate estimates are obtained by exponential filtering:

$$\tau_r \frac{dr_{\text{pre}}}{dt} = -r_{\text{pre}} + \rho_{\text{pre}(t)}, \quad \tau_r \frac{dr_{\text{post}}}{dt} = -r_{\text{post}} + \rho_{\text{post}(t)}. \tag{6}$$

2.4 Weight Update: The Three-Factor Rule

The weight evolves under the product of the eligibility trace and a global modulation signal $M(t)$:

$$\frac{dw}{dt} = M(t) \cdot E(t) \tag{7}$$

The modulation signal M is computed from a reward $R(t)$ and a slowly adapting baseline $\overline{R}(t)$. The specific form of R depends on the *reward regime* (see Section 3).

3 Reward Regimes

All regimes share the three-factor structure Equation 7. They differ only in how R, M, and the baseline target are computed.

3.1 Rate Ratio (default)

$$R = -\left(r_{\text{post}} - \alpha r_{\text{pre}}\right)^2, \quad M = R - \overline{R}, \quad \frac{d\overline{R}}{dt} = \frac{-\overline{R} + R}{\tau_{\overline{R}}} \tag{8}$$

The postsynaptic rate is penalised for deviating from a fraction α of the presynaptic rate. $M = R - \overline{R}$ is a *reward prediction error* (RPE), analogous to phasic dopamine (Schultz et al., 1997).

This is the R-STDP framework from Fremaux & Gerstner (Frémaux & Gerstner, 2016) (Eq. 7, Section 4.3).

3.2 BCM Sliding Threshold

$$R = r_{\text{post}}\big(r_{\text{post}} - \theta_M\big), \quad M = R, \quad \frac{d\theta_M}{dt} = \frac{r_{\text{post}}^2/r_{\text{target}} - \theta_M}{\tau_{\overline{R}}} \tag{9}$$

The sliding threshold θ_M tracks $\langle r_{\text{post}}^2\rangle/r_{\text{target}}$. At equilibrium $\theta_M = r^2/r_{\text{target}}$, giving $R = r^2\big(1 - r/r_{\text{target}}\big)$: LTP when $r < r_{\text{target}}$, LTD when $r > r_{\text{target}}$. This provides automatic homeostatic stabilisation.

Source: Bienenstock, Cooper & Munro (Bienenstock et al., 1982) — Eq. 4 (modification function φ), Eq. 7 (sliding threshold), Figure 2 (the φ curve).

3.3 Fixed Target Rate

$$R = -\big(r_{\text{post}} - r_{\text{target}}\big)^2, \quad M = R - \overline{R} \tag{10}$$

A simplified version of rate ratio with an absolute target (no dependence on r_{pre}). Models brain–computer interface paradigms where a neuron must learn a specific firing rate (Florian, 2007) (Section 4.4, "Learning a Target Firing-Rate Pattern").

3.4 Information Maximisation

$$R = \log\big(r_{\text{post}}/r_0\big) - \beta\big(r_{\text{post}} - r_0\big), \quad M = R - \overline{R} \tag{11}$$

Derived from the gradient of mutual information $I(X;Y)$ between input and output spike trains, subject to a firing-rate constraint (Toyoizumi et al., 2005). The log term encourages spiking (Fisher information); the linear penalty enforces metabolic cost. The balance produces a BCM-like learning rule (Toyoizumi et al., Eqs. 12 and 16, Figure 2).

3.5 Cross-Correlation (Temporal Precision)

$$R = x_{\text{pre}} y_{\text{post}} - C_{\text{target}}, \quad M = R - \overline{R} \tag{12}$$

Rewards temporal coincidence: R is large only when pre- and post-spikes occur within $\sim \tau_+$ of each other. With $C_{\text{target}} = 0$, any coincidence is rewarded. Relevant to auditory localisation and sequence learning (Gerstner et al., 2014) (Chapter 19).

3.6 Surprise / Novelty

$$R = \Big(r_{\text{post}} - \overline{R}\Big)^2, \quad M = R, \quad \frac{d\overline{R}}{dt} = \frac{-\overline{R} + r_{\text{post}}}{\tau_{\overline{R}}} \tag{13}$$

Here $\overline{R}$ tracks the *rate* (not the reward). Surprise $= \Big(r_{\text{post}} - \overline{R}\Big)^2$ is always non-negative; the eligibility trace E alone determines LTP vs LTD direction. Models noradrenergic/cholinergic modulation: novel stimuli enhance plasticity regardless of valence (Yu & Dayan, 2005) (Introduction, p. 681; Discussion, pp. 687–688).

3.7 ISI / Temporal Regularity

$$R = -\big(\big(r_{\text{post}}/r_{\text{target,ISI}}\big) - 1\big)^2, \quad M = R - \overline{R} \tag{14}$$

where $r_{\text{target,ISI}} = \tau_r/\text{ISI}_{\text{target}}$ converts the target interspike interval to rate-trace units. Penalises fractional deviation from a target ISI, relevant to burst coding and regularity regimes (Zeldenrust et al., 2018).

4 Complete System of Equations

State vector: $\boldsymbol{x}(t) = \big(V, y_{\text{post}}, E, r_{\text{post}}, \overline{R}, w\big) \in \mathbb{R}^6$.
Input signals: $\big(I_{s(t)}, x_{\text{pre}(t)}, r_{\text{pre}(t)}\big)$ from the presynaptic subsystem.

4.0.1 Inter-spike dynamics (between events)

$$\begin{aligned}
\tau_m \frac{dV}{dt} &= -(V - E_L) + R_m w I_{s(t)} \\
\frac{dy_{\text{post}}}{dt} &= -y_{\text{post}}/\tau_- \\
\frac{dE}{dt} &= -E/\tau_e \\
\frac{dr_{\text{post}}}{dt} &= -r_{\text{post}}/\tau_r \\
\tau_{\overline{R}} \frac{d\overline{R}}{dt} &= -\overline{R} + T_{\text{regime}} \\
\frac{dw}{dt} &= M \cdot E
\end{aligned} \tag{15}$$

where T_{regime} is the baseline target (R for most regimes, $r_{\text{post}}^2/r_{\text{target}}$ for BCM, r_{post} for surprise).

4.0.2 Pre-synaptic spike at t_{pre}^k

$$I_s \leftarrow I_s + 1, \quad x_{\text{pre}} \leftarrow x_{\text{pre}} + 1, \quad r_{\text{pre}} \leftarrow r_{\text{pre}} + 1 \tag{16}$$

$$E \leftarrow E - \eta_- w y_{\text{post}(t_{\text{pre}}^{k-})} \quad \text{(LTD eligibility)} \tag{17}$$

4.0.3 Post-synaptic spike at t_{post}^k

$$V \leftarrow V_{\text{reset}}, \quad y_{\text{post}} \leftarrow y_{\text{post}} + 1, \quad r_{\text{post}} \leftarrow r_{\text{post}} + 1 \tag{18}$$

$$E \leftarrow E + \eta_+ (w_{\max} - w) x_{\text{pre}(t_{\text{post}}^{k-})} \quad \text{(LTP eligibility)} \tag{19}$$

5 Numerical Methods

5.1 Hybrid Integration Scheme

The simulation uses a hybrid event-driven / continuous scheme. Smooth dynamics are integrated by Euler or classical RK4 between spike events. Discrete events (pre/post spikes) apply instantaneous jumps to the state vector.

For the RK4 path, threshold crossing detection uses linear interpolation: given V_0 and $V_1 = V(t + \Delta t)$ from a trial step, the crossing fraction is

$$f = \frac{\theta - V_0}{V_1 - V_0} \tag{20}$$

The timestep is then split: integrate to $t + f\Delta t$, apply the spike, integrate the remainder $t + f\Delta t \to t + \Delta t$. This is first-order accurate in spike time (error $\mathcal{O}(\Delta t^2)$).

5.2 RK4 Convergence

Convergence was measured against a fine-Δt reference solution ($\Delta t = 10\mu s$):

Variable	Euler (expect 1)	RK4 (expect 4)	Note
V (mV)	0.20	1.57	Degraded by spike discontinuities
w	1.79	1.25	Slow variable; averaging effects

Spike discontinuities (voltage resets, trace jumps) violate the smoothness assumption underlying classical convergence theorems. The linear interpolation for spike times limits accuracy to $\mathcal{O}(\Delta t^2)$, reducing RK4′s effective order. Euler benefits from compensating-error averaging on the slow weight variable.

6 Analysis Results

All simulations use $\Delta t = 0.1$ ms, RK4, $T = 10$ s unless noted.

6.1 Timescale Hierarchy

Variable	Dominant dynamics	Timescale
V	Spikes + subthreshold integration	$\tau_m = 20$ ms
I_s	Impulse + fast decay	$\tau_s = 5$ ms
$x_{\text{pre}}, y_{\text{post}}$	Impulse + moderate decay	$\tau_\pm = 20$ ms
E	STDP jumps + slow decay	$\tau_e = 500$ ms
$r_{\text{pre}}, r_{\text{post}}$	Impulse + slow decay	$\tau_r = 500$ ms
$\overline{R}$	Very slow tracking	$\tau_{\overline{R}} = 5$ s
w	Slow drift via $M \cdot E$	Emergent; seconds

6.2 Parameter Sensitivity

The most critical parameters are:

w_0 (initial weight) — bistability. A critical weight $w_{\text{crit}} \approx 1.5$–$2.0$ separates a silent regime from an active one. Below w_{crit}, the neuron never fires, $E = 0$ identically, and $dw/dt = M \cdot 0 = 0$ — the weight is frozen forever. This is the *silent synapse problem* (Frémaux & Gerstner, 2016) (Section 4.3): reward-modulated STDP cannot bootstrap a silent synapse.

The critical weight can be estimated from $V_{\text{ss}} \approx E_L + R_m w r_{\text{pre}} \tau_s$. Setting $V_{\text{ss}} = \theta$ gives $w_{\text{crit}} \approx \frac{\theta - E_L}{R_m r_{\text{pre}} \tau_s} \approx 3$, reduced to ~ 1.5 by peak-to-mean EPSP ratios.

R_m (membrane resistance) — high sensitivity. Directly scales synaptic drive; increasing R_m from 10 to 120 sweeps the post rate from 4 to 20 Hz.

α (target ratio) — moderate sensitivity. The weight adjusts to track the target direction but doesn't converge precisely within $T = 10$ s.

η_+, τ_e, $\tau_{\overline{R}}$ — low sensitivity. Robust across order-of-magnitude changes.

6.3 Regime Comparison

All 7 regimes were run with identical parameters ($w_0 = 2$, $r_{\text{pre}} = 20$ Hz, $T = 10$ s):

Regime	Post rate (Hz)	w_{final}	Behaviour
rate_ratio	13.4	1.92	Active regulation; ongoing M oscillations
bcm	20.0	4.50	Strong potentiation to BCM equilibrium
fixed_target	20.0	2.06	Minimal change; RPE cancellation
infomax	20.0	2.00	Minimal change; RPE cancellation
correlation	20.0	2.00	Zero-mean coincidence fluctuations
surprise	20.0	3.34	Monotonic potentiation; $M > 0$ always
isi_target	20.0	2.01	Minimal change; RPE cancellation

6.3.1 Key finding: RPE baseline cancellation

With deterministic pre-spikes, the post-neuron fires at a nearly constant rate, making R constant. Once $\overline{R}$ converges to R, the modulation $M = R - \overline{R} \to 0$ and learning stops — regardless of distance from the target. This affects all RPE-based regimes (fixed_target, infomax, correlation, isi_target).

Rate ratio avoids this because R depends on *both* r_{pre} and r_{post}, creating ongoing fluctuations that keep $M \neq 0$.

BCM and surprise avoid this because $M = R$ (no baseline subtraction), so $M \neq 0$ whenever $R \neq 0$.

In biological networks, Poisson spiking and network noise would provide the reward variability that RPE needs. The deterministic pre-spikes expose a structural limitation of RPE-based three-factor rules.

7 Timescale Separation and Relaxation Oscillations

The reward baseline $\overline{R}$ and weight w evolve on timescales much slower than all other state variables. The baseline has $\tau_{\overline{R}} = 5$ s (250 × slower than τ_m). The weight evolves through $\dot{w} = M \cdot E$ where both M and E (scaled by $\eta_\pm \sim 10^{-4}$) are small.

7.1 Spike-Count Quantisation

The postsynaptic neuron fires a discrete number of spikes $N(w)$ per presynaptic inter-spike interval. At critical weights $w_c^{(n)}$ where N increments from n to $n+1$, the firing rate r_{post} undergoes a step change, producing a discontinuous shift in R.

7.2 Mechanism of the Limit Cycle

Near $w_c^{(n)}$, the system exhibits a relaxation oscillation:

1. **Slow recovery.** With $w < w_c^{(n)}$, the neuron fires $N = n$ spikes. $M \approx 0$, $\dot{w} \approx 0$. Residual $M > 0$ drifts w upward.
2. **Threshold crossing.** When w crosses $w_c^{(n)}$, an extra spike fires. r_{post} jumps, R drops sharply. Since $\overline{R}$ hasn't tracked the change ($\tau_{\overline{R}}$ is large), $M = R - \overline{R} \ll 0$, driving $\dot{w} \ll 0$.
3. **Subcritical reset.** w is pushed below $w_c^{(n)}$, the extra spike ceases, R recovers, and the cycle repeats.

This is a *grazing bifurcation*: the smooth weight dynamics interact with a discontinuous reset map at the spike threshold, generically producing a stable limit cycle rather than a fixed point.

8 Parameters

Parameter	Description	Value	Source
τ_m	Membrane time constant	20 ms	(Dayan & Abbott, 2001)
E_L	Resting potential	-65 mV	(Dayan & Abbott, 2001)
R_m	Membrane resistance	50 MΩ	(Dayan & Abbott, 2001)
θ	Spike threshold	-50 mV	(Gerstner et al., 2014)
V_{reset}	Reset potential	-70 mV	(Gerstner et al., 2014)
τ_{ref}	Refractory period	3 ms	(Gerstner et al., 2014)
τ_s	Synaptic decay	5 ms	(Dayan & Abbott, 2001)
τ_+, τ_-	STDP windows	20 ms	(Bi & Poo, 1998)
τ_r	Rate trace decay	500 ms	(Gerstner et al., 2014)
τ_e	Eligibility decay	500 ms	(Frémaux & Gerstner, 2016)
$\tau_{\overline{R}}$	Reward baseline	5 s	(Frémaux & Gerstner, 2016)
α	Target rate ratio	0.5	—
η_+, η_-	Learning rates	10^{-4}	(Song et al., 2000)
$w_{\max}$	Max weight	10	—
w_0	Initial weight	2.0	—

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